Character Theory

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Contents

1 Introduction **1**
1.1 Characters . . . 1
1.2 Schur's lemma . . . 3

2 Orthogonality relations for characters **3**

3 Decomposition of the regular representation **7**

4 Number of irreducible representations **8**

5 Induced representations **14**
5.1 Restriction and the construction of induction . . . 14
5.2 The induced character . . . 16
5.3 Frobenius reciprocity . . . 17
5.4 A calculation in S_3 . . . 18
5.5 Application: a group of order 21 . . . 19

6 Arithmetic properties of characters **21**
6.1 Algebraic integers . . . 21
6.2 Class sums and character degrees . . . 22
6.3 Applications to character degrees . . . 23

1 Introduction

1.1 Characters

Let G be a finite group, let $\rho\colon G \to \mathrm{GL}(V)$ be a representation. The *character* of ρ is defined as

$$\chi_\rho(s) = \mathrm{Tr}(\rho_s).$$

Example 1.1. Let $G = S_2 = \{1, s\}$, where $s = (12)$, act on $\mathbb{C}^2$ by letting 1 leave both coordinates unchanged and s swap them:

$$\rho_1(x, y) = (x, y), \qquad \rho_s(x, y) = (y, x).$$

Swapping twice gives the identity, so this defines a representation of G. In the standard basis, its matrices are

$$\rho_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \rho_s = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The character assigns to each group element the sum of the diagonal entries of its matrix:

$$\chi_\rho(1) = 1 + 1 = 2, \qquad \chi_\rho(s) = 0 + 0 = 0.$$

NOTE Here we have

$$\chi_\rho(s^2) = \chi_\rho(1) = 2 \neq \chi_\rho(s)^2.$$

Example 1.2. There is a natural representation of S_3 on $\mathbb{C}^3$ by permuting the standard basis:

$$\sigma(e_i) = e_{\sigma(i)}.$$

The standard representation is the restriction to the subspace

$$W := \{x_1 + x_2 + x_3 = 0\}.$$

For explicit matrices, choose the basis $u = e_1 - e_2$, $v = e_2 - e_3$. Then

$$\rho((12)) = \begin{pmatrix} -1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \rho((123)) = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

These two permutations generate S_3, so these matrices determine the representation.

Its character is

$$\chi_{\text{std}}((1)) = 2, \qquad \chi_{\text{std}}((12)) = 0, \qquad \chi_{\text{std}}((123)) = -1.$$

Proposition 1.3. *If χ is the character of a representation ρ of degree n, we have:*

(i) $\chi(1) = n$,

(ii) $\chi(s^{-1}) = \overline{\chi(s)}$,

(iii) $\chi(tst^{-1}) = \chi(s)$.

Proof. (2) This is because ρ_s has finite order and thus its eigenvalues $\lambda_1, \ldots, \lambda_n$ have absolute value equal to 1.

(3) This follows from $\chi(vu) = \chi(uv)$ by putting $u = ts$, $v = t^{-1}$. □

Proposition 1.4. *Let $\rho^1 \colon G \to \mathrm{GL}(V_1)$ and $\rho^2 \colon G \to \mathrm{GL}(V_2)$ be two representations.*

(i) *The character χ of the direct sum representation $V_1 \oplus V_2$ is equal to $\chi_1 + \chi_2$.*

(ii) *The character ψ of the tensor product representation $V_1 \otimes V_2$ is equal to $\chi_1 \cdot \chi_2$.*

Sketch of (ii). Assume $\dim V_1 = \dim V_2 = 2$, and choose bases e_1, e_2 of V_1 and f_1, f_2 of V_2. Fix $s \in G$, and write $A = (a_{ij})$ and $B = (b_{ij})$ for the matrices of ρ^1_s and ρ^2_s in these bases. Since s acts on $e_i \otimes f_j$ by $\rho^1_s(e_i) \otimes \rho^2_s(f_j)$, its matrix in the ordered basis $e_1 \otimes f_1, e_1 \otimes f_2, e_2 \otimes f_1, e_2 \otimes f_2$ is

$$A \otimes B = \begin{pmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{pmatrix}.$$

Taking the trace gives

$$\psi(s) = a_{11}b_{11} + a_{11}b_{22} + a_{22}b_{11} + a_{22}b_{22} = (a_{11} + a_{22})(b_{11} + b_{22}) = \chi_1(s)\chi_2(s).$$

□

1.2 Schur's lemma

Theorem 1.5 (Maschke's lemma)**.** *Let $\rho\colon G \to \mathrm{GL}(V)$ be a linear representation of G in V and let W be a vector subspace of V stable under G. Then there exists a complement W^0 of W in V which is stable under G.*

Proof. Let W' be an arbitrary complement of W in V, and let p be the corresponding projection of V onto W. Form the average p^0 of the conjugates of p by the elements of G:

$$p^0 = \frac{1}{g}\sum_{t\in G} \rho_t \cdot p \cdot \rho_t^{-1} \qquad (g \text{ being the order of } G).$$

For $x \in W$ we have $p^0 x = x$. Thus p^0 is a projection of V onto W, corresponding to some complement W^0 of W. One can show that W^0 is table under G. □

Proposition 1.6 (Schur's lemma)**.** *Let $\rho^1\colon G \to \mathrm{GL}(V_1)$ and $\rho^2\colon G \to \mathrm{GL}(V_2)$ be two irreducible representations of G, and let f be a linear mapping of V_1 into V_2 such that $\rho^2_s \circ f = f \circ \rho^1_s$ for all $s \in G$. Then*

(1) *If ρ^1 and ρ^2 are not isomorphic, we have $f = 0$.*

(2) *If $V_1 = V_2$ and $\rho^1 = \rho^2$, f is a homothety (i.e., a scalar multiple of the identity).*

Sketch. (1) The intertwining identity shows that $\ker f$ and $\operatorname{im} f$ are G-invariant. If $f \neq 0$, irreducibility gives $\ker f = 0$ and $\operatorname{im} f = V_2$, so f is an isomorphism of representations. Hence $f = 0$ when ρ^1 and ρ^2 are not isomorphic.

(2) Set $V = V_1 = V_2$ and $\rho = \rho^1 = \rho^2$. Since V is a nonzero finite-dimensional complex vector space, f has an eigenvalue λ. For $v \in E_\lambda := \ker(f - \lambda I_V)$ and $s \in G$, we have

$$f(\rho_s v) = \rho_s f(v) = \lambda \rho_s v.$$

Thus E_λ is a nonzero G-invariant subspace, so irreducibility gives $E_\lambda = V$ and $f = \lambda I_V$. □

2 Orthogonality relations for characters

Definition 2.1. Let V be a vector space over $\mathbb{R}$ or $\mathbb{C}$. An *inner product* on V is a map $\langle\ ,\ \rangle\colon V \times V \to \mathbb{R}$ or $\mathbb{C}$, respectively, satisfying the following conditions for all $u, v, w \in V$ and scalars a, b:

(i) $\langle au + bv, w\rangle = a\langle u, w\rangle + b\langle v, w\rangle$;

(ii) $\langle u, v\rangle = \overline{\langle v, u\rangle}$;

(iii) $\langle v, v\rangle$ is real and nonnegative, and equals zero if and only if $v = 0$.

Thus the inner product is conjugate-linear in its second variable:

$$\langle u, av + bw\rangle = \overline{a}\langle u, v\rangle + \overline{b}\langle u, w\rangle.$$

Vectors u, v are *orthogonal* if $\langle u, v\rangle = 0$.

Example 2.2. For a finite group G, the vector space of functions $f\colon G \to \mathbb{C}$ has the inner product

$$\langle f, h\rangle = \frac{1}{|G|}\sum_{s\in G} f(s)\overline{h(s)}.$$

Linearity and conjugate symmetry follow from the formula, and positive definiteness follows from $\langle f, f\rangle = \frac{1}{|G|}\sum_{s\in G} |f(s)|^2$. In particular, this inner product applies to characters.

Definition 2.3. Let $\rho^i\colon G \to \mathrm{GL}(V_i)$ $(i = 1, 2)$ be two representations with character ϕ and ψ. Define

$$\langle \varphi, \psi \rangle = \frac{1}{|G|} \sum_{t \in G} \phi(t)\overline{\psi(t)}.$$

Exercise 2.1. Calculate $\langle \chi_{\mathrm{std}}, \chi_{\mathrm{std}} \rangle$ for the standard representation of S_3 in Example 1.2.

Theorem 2.4. (i) *If χ is the character of an irreducible representation, we have $\langle \chi, \chi \rangle = 1$.*

(ii) *If χ and χ' are the characters of two nonisomorphic irreducible representations, we have $\langle \chi, \chi' \rangle = 0$.*

Proof. Let $\rho\colon G \to \mathrm{GL}(V)$ and $\sigma\colon G \to \mathrm{GL}(W)$ be finite-dimensional complex representations with characters χ and ψ. We first prove

$$\langle \chi, \psi \rangle = \dim \mathrm{Hom}_G(W, V).$$

Here $\mathrm{Hom}_G(W, V)$ is the space of linear maps $F\colon W \to V$ for which the following diagram commutes for every $g \in G$:

$$\begin{array}{ccc} W & \xrightarrow{F} & V \\ {\scriptstyle \sigma(g)}\downarrow & & \downarrow{\scriptstyle \rho(g)} \\ W & \xrightarrow{F} & V \end{array}$$

Equivalently, $\rho(g) \circ F = F \circ \sigma(g)$ for every $g \in G$.

Let $H := \mathrm{Hom}_{\mathbb{C}}(W, V)$. Define the averaging map

$$\begin{aligned} P\colon \mathrm{Hom}_{\mathbb{C}}(W, V) &\longrightarrow \mathrm{Hom}_G(W, V), \\ F &\longmapsto \frac{1}{|G|} \sum_{g \in G} \rho(g) F \sigma(g)^{-1}. \end{aligned}$$

To see that $P(F)$ belongs to $\mathrm{Hom}_G(W, V)$, we calculate for every $h \in G$ that

$$\rho(h)P(F)\sigma(h)^{-1} = \frac{1}{|G|} \sum_{g \in G} \rho(hg) F \sigma(hg)^{-1} = P(F).$$

Moreover, P fixes $\mathrm{Hom}_G(W, V)$. Via the inclusion $\mathrm{Hom}_G(W, V) \subseteq H$, we also regard P as an endomorphism of H. Therefore P is a projection onto $\mathrm{Hom}_G(W, V)$. Since $H = \operatorname{im} P \oplus \ker P$ and P acts as the identity on its image and as zero on its kernel, we obtain

$$\mathrm{Tr}_H(P) = \dim \mathrm{Hom}_G(W, V).$$

Here $\mathrm{Tr}_H(P)$ denotes the trace of P as a linear operator on H.

We now compute this trace using characters. Choose bases of V and W, and identify H with the space of $n \times m$ matrices, where $n = \dim V$ and $m = \dim W$. For fixed $n \times n$ and $m \times m$ matrices A and B, respectively, the operator $F \mapsto AFB$ on H has trace

$$\mathrm{Tr}_H(F \mapsto AFB) = \sum_{i=1}^{n} \sum_{j=1}^{m} A_{ii} B_{jj} = \mathrm{Tr}(A)\,\mathrm{Tr}(B).$$

Indeed, in the matrix-unit basis E_{ij}, the coefficient of E_{ij} in $AE_{ij}B$ is $A_{ii}B_{jj}$. Applying this identity to each summand defining P gives

$$\begin{aligned}\mathrm{Tr}_H(P) &= \frac{1}{|G|}\sum_{g\in G}\mathrm{Tr}(\rho(g))\,\mathrm{Tr}(\sigma(g)^{-1}) \\ &= \frac{1}{|G|}\sum_{g\in G}\chi(g)\psi(g^{-1}) \\ &= \frac{1}{|G|}\sum_{g\in G}\chi(g)\overline{\psi(g)} = \langle\chi,\psi\rangle,\end{aligned}$$

where we used $\psi(g^{-1}) = \overline{\psi(g)}$. This proves the claimed identity.

For (i), take $W = V$ and $\sigma = \rho$ irreducible. By Schur's lemma, $\mathrm{Hom}_G(V,V)$ consists exactly of the scalar multiples of the identity, so it has dimension 1 and $\langle\chi,\chi\rangle = 1$. For (ii), take V and W to be nonisomorphic irreducible representations, with $\psi = \chi'$. Schur's lemma gives $\mathrm{Hom}_G(W,V) = 0$, hence $\langle\chi,\chi'\rangle = 0$. □

Theorem 2.5. *Let V be a linear representation of G, with character ϕ, and suppose V decomposes into a direct sum of irreducible representations:*

$$V = W_1 \oplus \cdots \oplus W_k.$$

Then, if W is an irreducible representations with character χ, the number of W_i isomorphic to W is equal to the scalar product $\langle\phi,\chi\rangle$.

Proof. Let χ_i be the character of W_i. We have

$$\phi = \chi_1 + \cdots + \chi_k.$$

Then $\langle\phi,\chi\rangle = \langle\chi_1,\chi\rangle + \cdots + \langle\chi_k,\chi\rangle$. □

Corollary 2.6. *Two representations with the same character are isomorphic.*

Proof. Let V and V' have the same character ϕ. By Maschke's theorem, both decompose into direct sums of irreducible representations. For any irreducible representation W with character χ, the preceding theorem gives the same multiplicity $\langle\phi,\chi\rangle$ in both decompositions. Hence V and V' have the same irreducible summands with the same multiplicities, so $V \cong V'$ as representations of G. □

Example 2.7 (Orthogonality in S_3)**.** The trivial, sign, and standard characters of S_3 have the following values:

Conjugacy-class representative	1	(12)	(123)
Class size	1	3	2
χ_{triv}	1	1	1
χ_{sgn}	1	−1	1
χ_{std}	2	0	−1

Grouping the terms in each inner product by conjugacy class, we obtain

$$\begin{aligned}
\langle \chi_{\text{triv}}, \chi_{\text{triv}} \rangle &= \langle \chi_{\text{sgn}}, \chi_{\text{sgn}} \rangle = \frac{1+3+2}{6} = 1, \\
\langle \chi_{\text{std}}, \chi_{\text{std}} \rangle &= \frac{4+3\cdot 0+2\cdot 1}{6} = 1, \\
\langle \chi_{\text{triv}}, \chi_{\text{sgn}} \rangle &= \frac{1-3+2}{6} = 0, \\
\langle \chi_{\text{triv}}, \chi_{\text{std}} \rangle &= \langle \chi_{\text{sgn}}, \chi_{\text{std}} \rangle = \frac{2-2}{6} = 0.
\end{aligned}$$

So these three characters form an orthonormal set.

Question. Does S_3 have any irreducible complex characters other than the above three? More generally, given a finite group G, how can we determine the number of its distinct irreducible characters?

Exercise 2.2. A representation V of S_3 has character

$$\phi(1) = 7, \qquad \phi((12)) = 1, \qquad \phi((123)) = 1.$$

Determine its decomposition into irreducible representations. Calculate $\langle \phi, \phi \rangle$, and check that the dimensions of the summands add up correctly.

Exercise 2.3. Let $G = \langle g \mid g^3 = 1 \rangle$ and $\omega = e^{2\pi i/3}$. For $j = 0, 1, 2$, define

$$\chi_j(g^k) = \omega^{jk}.$$

Show that these are characters of one-dimensional representations and prove

$$\langle \chi_i, \chi_j \rangle = \delta_{ij}.$$

Calculate $\frac{1}{3}\sum_{k=0}^{2} \chi_1(g^k)^2$ and explain why complex conjugation is necessary in the inner product.

Example 2.8 (The character table of D_8)**.** Let $G = D_8$ be the symmetry group of a square, of order 8, with presentation

$$G = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle,$$

where r is rotation through 90° and s is a reflection. Conjugating by r and s gives the five conjugacy classes

$$\{1\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, r^2 s\}, \quad \{rs, r^3 s\}.$$

For each $\varepsilon, \delta \in \{1, -1\}$, the assignments

$$\lambda_{\varepsilon,\delta}(r) = \varepsilon, \qquad \lambda_{\varepsilon,\delta}(s) = \delta$$

satisfy the defining relations and give a one-dimensional representation, whose character we also denote by $\lambda_{\varepsilon,\delta}$. There is also a representation $\rho\colon G \to \mathrm{GL}_2(\mathbb{C})$ defined by

$$\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \rho(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

These matrices satisfy the defining relations. Taking traces gives the following values, where χ is the character of ρ and the signs in the subscripts stand for 1 and -1:

Conjugacy-class representative	1	r^2	r	s	rs
Class size	1	1	2	2	2
$\lambda_{+,+}$	1	1	1	1	1
$\lambda_{+,-}$	1	1	1	-1	-1
$\lambda_{-,+}$	1	1	-1	1	-1
$\lambda_{-,-}$	1	1	-1	-1	1
χ	2	-2	0	0	0

The four one-dimensional representations are irreducible, and

$$\langle \chi, \chi \rangle = \frac{2^2 + (-2)^2}{8} = 1.$$

3 Decomposition of the regular representation

Definition 3.1 (Regular representation)**.** Let G be a finite group, and let

$$\mathbb{C}[G] = \bigoplus_{t \in G} \mathbb{C} e_t$$

be the vector space with basis $\{e_t : t \in G\}$. The *left regular representation* of G on $\mathbb{C}[G]$ is defined by

$$\rho_{\mathrm{reg}}(s)(e_t) = e_{st}, \qquad s, t \in G,$$

extended linearly to all of $\mathbb{C}[G]$. This is a representation of degree $|G|$.

Exercise 3.1. Compute the character χ_{reg} of the left regular representation ρ_{reg}.

Answer.

$$\chi_{\mathrm{reg}}(g) = \begin{cases} |G|, & g = 1, \\ 0, & g \neq 1. \end{cases}$$

Proposition 3.2. *Every irreducible complex representation W_i of G occurs in the left regular representation with multiplicity* $\dim W_i$*. In particular, $\mathbb{C}[G]$ contains a subrepresentation isomorphic to W_i.*

Proof. By Maschke's theorem, $\mathbb{C}[G]$ decomposes into a direct sum of irreducible representations. Let χ_i be the character of W_i. We have

$$\begin{aligned} n_i &= \langle \chi_{\mathrm{reg}}, \chi_i \rangle \\ &= \frac{1}{|G|} \sum_{g \in G} \chi_{\mathrm{reg}}(g) \overline{\chi_i(g)} \\ &= \frac{1}{|G|} |G| \overline{\chi_i(1)} = \dim W_i. \end{aligned}$$

So W_i occurs as a direct summand of $\mathbb{C}[G]$ up to isomorphism. □

Corollary 3.3. (a) *The degrees n_i satisfy the relation $\sum_{i=1}^{h} n_i^2 = g$.*

(b) *If $s \in G$ is different from 1, we have $\sum_{i=1}^{h} n_i \chi_i(s) = 0$.*

Proof. By Proposition 3.2, we have

$$\chi_{\text{reg}}(g) = \sum n_i \chi_i(g) \qquad \text{for all } g \in G.$$

Taking $g = 1$ we obtain (a), and taking $s \neq 1$, we obtain (b). □

Remark 3.4. If $W_1, \ldots, W_k$ be all the irreducible representations of G, mutually nonisomorphic, with degree n_i, then $n_1^2 + \cdots + n_k^2 = g$.

Example 3.5 (All irreducible representations of S_3)**.** By the multiplicity formula, the distinct characters of norm 1 in Example 2.7 correspond to pairwise nonisomorphic irreducible representations. Since their degrees satisfy

$$1^2 + 1^2 + 2^2 = 6 = |S_3|,$$

the sum-of-squares formula shows that these are all the irreducible complex representations of S_3, up to isomorphism.

Exercise 3.2 (Determining irreducible degrees)**.** Suppose $|G| = 60$ and G has exactly five irreducible complex representations up to isomorphism, only one of which is one-dimensional. Determine their degrees and decompose $\mathbb{C}[G]$.

Answer. The four nontrivial degrees have squares summing to 59. Modulo 4 and 8, exactly one is even and divisible by 4; its square is at most $59 - 3 \cdot 9 = 32$, so this degree is 4. The remaining odd degrees have squares summing to 43, forcing $3, 3, 5$. Labeling the irreducibles $W_1, \ldots, W_5$ with degrees $1, 3, 3, 4, 5$, respectively, we obtain

$$\mathbb{C}[G] \cong W_1 \oplus W_2^{\oplus 3} \oplus W_3^{\oplus 3} \oplus W_4^{\oplus 4} \oplus W_5^{\oplus 5}.$$

4 Number of irreducible representations

A function $f : G \to \mathbb{C}$ is called a *class function* if

$$f(tst^{-1}) = f(s)$$

for all $s, t \in G$. Denote by H the space of class functions on G. Let $\chi_1, \ldots, \chi_h$ be the characters of a complete set of pairwise nonisomorphic irreducible complex representations of G.

Theorem 4.1. *The characters $\chi_1, \ldots, \chi_h$ form an orthonormal basis of H.*

Proof. The characters are orthonormal class functions, so it remains to show that they span H. Let $u \in H$ and set

$$f = u - \sum_{i=1}^{h} \langle u, \chi_i \rangle \chi_i.$$

Orthonormality gives

$$\langle f, \chi_j \rangle = \langle u, \chi_j \rangle - \langle u, \chi_j \rangle = 0 \qquad (1 \leq j \leq h).$$

It suffices to show that $f = 0$. On the regular representation $\mathbb{C}[G]$, define

$$T = \frac{1}{|G|} \sum_{s \in G} f(s) \rho_{\text{reg}}(s^{-1}).$$

For every $t \in G$, conjugating T and then reindexing by $x = tst^{-1}$ gives

$$\begin{aligned}\rho_{\mathrm{reg}}(t)T\rho_{\mathrm{reg}}(t)^{-1} &= \frac{1}{|G|}\sum_{s\in G} f(s)\rho_{\mathrm{reg}}(ts^{-1}t^{-1}) \\ &= \frac{1}{|G|}\sum_{x\in G} f(t^{-1}xt)\rho_{\mathrm{reg}}(x^{-1}) \\ &= T.\end{aligned}$$

By Maschke's theorem, $\mathbb{C}[G]$ is a direct sum of irreducible subrepresentations. Since T is a linear combination of the operators $\rho_{\mathrm{reg}}(s)$, it preserves each summand. On any such summand U with character χ_U, Schur's lemma gives $T|_U = \lambda I_U$, and

$$\lambda \dim U = \mathrm{Tr}(T|_U) = \frac{1}{|G|}\sum_{s\in G} f(s)\chi_U(s^{-1}) = \langle f, \chi_U\rangle = 0.$$

Thus $\lambda = 0$ on every summand, so $T = 0$. Applying T to the basis vector e_1 yields

$$0 = T(e_1) = \frac{1}{|G|}\sum_{s\in G} f(s)e_{s^{-1}}.$$

The vectors $e_{s^{-1}}$ form a basis of $\mathbb{C}[G]$, hence $f = 0$. Thus $u = \sum_{i=1}^{h}\langle u, \chi_i\rangle\chi_i$, proving that the characters span H. □

Theorem 4.2. *The number of irreducible representations of G (up to isomorphism) is equal to the number of conjugacy classes of G.*

Proof. Let $C_1, \ldots, C_r$ be the conjugacy classes of G, and let δ_j be 1 on C_j and 0 elsewhere. A class function is uniquely determined by its values on these classes, so $\delta_1, \ldots, \delta_r$ form a basis of H. Therefore $r = \dim H = h$ by Theorem 4.1. □

Theorem 4.3 (Column orthogonality)**.** *Let $C_1, \ldots, C_r$ be the conjugacy classes of G, and choose representatives $c_j \in C_j$. For $1 \le j, k \le r$, we have*

$$\sum_{i=1}^{h}\chi_i(c_j)\overline{\chi_i(c_k)} = \begin{cases}\dfrac{|G|}{|C_j|} = |C_G(c_j)|, & j = k, \\ 0, & j \neq k.\end{cases}$$

Here $C_G(c_j) = \{g \in G : gc_j = c_jg\}$ is the centralizer of c_j.

Proof. Fix k, and let δ_k be the class function equal to 1 on C_k and 0 elsewhere. Since each χ_i is constant on C_k, we have

$$\langle \delta_k, \chi_i\rangle = \frac{1}{|G|}\sum_{s\in C_k}\overline{\chi_i(s)} = \frac{|C_k|}{|G|}\overline{\chi_i(c_k)}.$$

Expanding δ_k in the orthonormal basis from Theorem 4.1 gives

$$\delta_k = \sum_{i=1}^{h}\langle \delta_k, \chi_i\rangle\chi_i = \frac{|C_k|}{|G|}\sum_{i=1}^{h}\overline{\chi_i(c_k)}\chi_i.$$

Evaluating at c_j, we obtain

$$\delta_{jk} = \frac{|C_k|}{|G|} \sum_{i=1}^{h} \chi_i(c_j)\overline{\chi_i(c_k)}.$$

This gives the stated values, and the equality $|G|/|C_j| = |C_G(c_j)|$ follows from the conjugacy-class size formula. □

Exercise 4.1 (Counting irreducible representations)**.** For C_4 (the cyclic group of order 4), list the conjugacy classes and determine the number of irreducible complex representations up to isomorphism.

Exercise 4.2 (Finite abelian groups)**.** Let G be a finite abelian group of order n. Show that G has exactly n irreducible complex representations up to isomorphism, all of degree 1. *Hint:* Combine the number of conjugacy classes with the sum-of-squares formula.

Example 4.4 (Characters of S_4)**.** Determine the conjugacy classes of S_4 and their sizes, and hence the number of irreducible characters. Compute the complete character table. *Hint:* Let $W = \{(x_1, x_2, x_3, x_4) \in \mathbb{C}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$ with the action by permuting coordinates. Use its character, its product with the sign character, and column orthogonality.

Solution. Two permutations in S_4 are conjugate if and only if they have the same cycle type. The possible cycle types are 1^4, $2\,1^2$, 2^2, $3\,1$, and 4. There are $\binom{4}{2} = 6$ transpositions. A product of two disjoint transpositions is determined by a partition of the four letters into two unordered pairs, giving $\frac{1}{2}\binom{4}{2} = 3$ possibilities. A 3-cycle is determined by its three letters and one of the two cyclic orders on them, giving $\binom{4}{3} \cdot 2 = 8$ possibilities. Finally, there are $(4-1)! = 6$ distinct 4-cycles. Thus the conjugacy classes have the following representatives and sizes:

Representative	1	(12)	(12)(34)	(123)	(1234)
Class size	1	6	3	8	6

Since there are five conjugacy classes, there are exactly five irreducible complex characters.

The trivial character χ_{triv} takes the value 1 everywhere. The sign character χ_{sgn} takes the values

$$(1, -1, 1, 1, -1)$$

in the displayed order, because a k-cycle has sign $(-1)^{k-1}$. These two characters come from distinct one-dimensional representations and are therefore irreducible.

Let π be the character of the permutation representation on $\mathbb{C}^4$. A permutation matrix has a diagonal entry 1 precisely at each fixed letter, so $\pi(\sigma)$ is the number of letters fixed by σ. The line $L = \mathbb{C}(1, 1, 1, 1)$ is fixed pointwise, and

$$\mathbb{C}^4 = L \oplus W$$

as representations of S_4. Thus $\pi = \chi_{\text{triv}} + \chi_W$, and the values of χ_W are computed as follows:

σ	1	(12)	(12)(34)	(123)	(1234)
$\pi(\sigma)$	4	2	0	1	0
$\chi_W(\sigma) = \pi(\sigma) - 1$	3	1	-1	0	-1

To show that χ_W is irreducible, we calculate

$$\langle \chi_W, \chi_W \rangle = \frac{1 \cdot 3^2 + 6 \cdot 1^2 + 3 \cdot (-1)^2 + 8 \cdot 0^2 + 6 \cdot (-1)^2}{24} = \frac{9 + 6 + 3 + 0 + 6}{24} = 1.$$

The tensor product $W \otimes \mathrm{sgn}$ has character

$$\chi_{W \otimes \mathrm{sgn}} = \chi_W \chi_{\mathrm{sgn}} = (3, -1, -1, 0, 1).$$

Since $|\chi_{\mathrm{sgn}}(\sigma)| = 1$ for every σ, this character also has squared norm 1, so $W \otimes \mathrm{sgn}$ is irreducible. Its value at (12) is -1, whereas $\chi_W((12)) = 1$, so the two degree-3 representations are not isomorphic. We have therefore found four distinct irreducible characters, of degrees $1, 1, 3, 3$.

Denote the remaining irreducible character by ψ, and let $d = \psi(1)$ be its degree. Column orthogonality at the identity (Theorem 4.3) gives

$$1^2 + 1^2 + 3^2 + 3^2 + d^2 = 24,$$

so $d^2 = 4$ and $d = 2$. For $\sigma \neq 1$, the σ-column is orthogonal to the identity column. Since the entries of the identity column are the real degrees $1, 1, 3, 3, 2$, we obtain

$$0 = \chi_{\mathrm{triv}}(\sigma) + \chi_{\mathrm{sgn}}(\sigma) + 3\chi_W(\sigma) + 3\chi_{W \otimes \mathrm{sgn}}(\sigma) + 2\psi(\sigma).$$

Substituting the four nonidentity representatives yields

$$\begin{aligned}
\psi((12)) &= -\frac{1 - 1 + 3 - 3}{2} = 0, \\
\psi((12)(34)) &= -\frac{1 + 1 - 3 - 3}{2} = 2, \\
\psi((123)) &= -\frac{1 + 1 + 0 + 0}{2} = -1, \\
\psi((1234)) &= -\frac{1 - 1 - 3 + 3}{2} = 0.
\end{aligned}$$

Consequently, the complete character table is

Representative	1	(12)	$(12)(34)$	(123)	(1234)
Class size	1	6	3	8	6
χ_{triv}	1	1	1	1	1
χ_{sgn}	1	-1	1	1	-1
χ_W	3	1	-1	0	-1
$\chi_{W \otimes \mathrm{sgn}}$	3	-1	-1	0	1
ψ	2	0	2	-1	0

As a check on the last row, its squared norm is

$$\langle \psi, \psi \rangle = \frac{1 \cdot 2^2 + 6 \cdot 0^2 + 3 \cdot 2^2 + 8 \cdot (-1)^2 + 6 \cdot 0^2}{24} = \frac{4 + 12 + 8}{24} = 1.$$

□

Example 4.5 (Characters of the quaternion group)**.** Let

$$Q_8 = \langle a, b \mid a^4 = 1,\ a^2 = b^2,\ b^{-1}ab = a^{-1} \rangle.$$

Its eight elements are $1, a, a^2, a^3, b, ab, a^2b, a^3b$. In the usual quaternion notation, $a = i$, $b = j$, $ab = k$, and $a^2 = -1$. Compute the complete character table of Q_8.

Solution. The subgroup $Z = \{1, a^2\}$ is central, and $Q_8/Z \cong C_2 \times C_2$ is abelian. Thus each conjugacy class is contained in a coset of Z. Conjugation by b exchanges a and a^3, while conjugation by a exchanges b and a^2b, and exchanges ab and a^3b. Hence the conjugacy classes are

$$\{1\}, \quad \{a^2\}, \quad \{a, a^3\}, \quad \{b, a^2b\}, \quad \{ab, a^3b\},$$

of sizes $1, 1, 2, 2, 2$. There are therefore five irreducible complex characters.

The four characters of $Q_8/Z \cong C_2 \times C_2$ give four distinct one-dimensional characters of Q_8. Explicitly, for $\varepsilon, \delta \in \{1, -1\}$, define

$$\lambda_{\varepsilon,\delta}(a^r b^s) = \varepsilon^r \delta^s \qquad (0 \le r \le 3,\ 0 \le s \le 1).$$

The assignments $a \mapsto \varepsilon$ and $b \mapsto \delta$ satisfy the defining relations, so each $\lambda_{\varepsilon,\delta}$ is indeed a character. On the five class representatives $1, a^2, a, b, ab$, its values are

$$(1, 1, \varepsilon, \delta, \varepsilon\delta).$$

Let ψ be the remaining irreducible character, of degree d. The sum-of-squares formula gives

$$4 \cdot 1^2 + d^2 = 8, \qquad \text{so } d = 2.$$

By column orthogonality with the identity column, for $g \neq 1$ we have

$$0 = \sum_{\varepsilon,\delta \in \{1,-1\}} \lambda_{\varepsilon,\delta}(g) + 2\psi(g).$$

At $g = a^2$, all four one-dimensional characters have value 1, giving $\psi(a^2) = -2$. At a, b, and ab, their values sum to zero, giving

$$\psi(a) = \psi(b) = \psi(ab) = 0.$$

Consequently, the complete character table is

Representative	1	a^2	a	b	ab
Class size	1	1	2	2	2
$\lambda_{1,1}$	1	1	1	1	1
$\lambda_{1,-1}$	1	1	1	-1	-1
$\lambda_{-1,1}$	1	1	-1	1	-1
$\lambda_{-1,-1}$	1	1	-1	-1	1
ψ	2	-2	0	0	0

As a check, the last row has squared norm

$$\langle \psi, \psi \rangle = \frac{2^2 + (-2)^2}{8} = 1.$$

□

Example 4.6 (The character table of A_4 by orthogonality)**.** Compute the complete character table of A_4.

Solution. The elements of A_4 are the identity, three products of two disjoint transpositions, and eight 3-cycles, so $|A_4| = 12$. Put

$$d_1 = (12)(34), \qquad d_2 = (13)(24), \qquad d_3 = (14)(23).$$

Each d_i has order 2, and $d_1 d_2 = d_2 d_1 = d_3$, so $V_4 = \{1, d_1, d_2, d_3\}$ is a subgroup. Conjugation preserves cycle type, hence V_4 is normal in A_4 and $A_4/V_4 \cong C_3$.

Set $a = (123)$. The identities $a d_1 a^{-1} = d_3$ and $a d_3 a^{-1} = d_2$ show that $V_4 \backslash \{1\}$ is one conjugacy class. A permutation commuting with a must fix 4 and act on $\{1, 2, 3\}$ as a power of a. Thus $C_{A_4}(a) = \langle a \rangle$, and the conjugacy class of a has size $12/3 = 4$. Since A_4/V_4 is abelian, each conjugate of a lies in aV_4. This coset also has four elements, so it is precisely the conjugacy class of a. Similarly, the conjugacy class of a^{-1} is $a^{-1}V_4$; the two cosets are distinct because aV_4 has order 3 in the quotient. Hence the conjugacy classes and their sizes are

Class	$\{1\}$	$V_4 \backslash \{1\}$	aV_4	a^2V_4
Representative	1	d_1	a	a^2
Class size	1	3	4	4

By Section 4, there are exactly four irreducible complex characters.

Let $\omega = e^{2\pi i/3}$. For $j = 0, 1, 2$, the character of A_4/V_4 sending aV_4 to ω^j gives a one-dimensional character of A_4:

$$\lambda_j(a^m v) = \omega^{jm} \qquad (m \in \{0, 1, 2\},\ v \in V_4).$$

The exponent m is uniquely determined by the coset, and multiplication of cosets adds exponents modulo 3, so this formula is well-defined and multiplicative. These characters are irreducible because they have degree 1, and they are distinct because their values at a are distinct. Their values, in the displayed order of classes, are

$$\lambda_0 = (1, 1, 1, 1), \qquad \lambda_1 = (1, 1, \omega, \omega^2), \qquad \lambda_2 = (1, 1, \omega^2, \omega).$$

Let χ be the remaining irreducible character and put $d = \chi(1)$. The sum-of-squares formula from Section 3 gives

$$1^2 + 1^2 + 1^2 + d^2 = 12, \qquad \text{hence } d = 3.$$

By column orthogonality with the identity column (Theorem 4.3), for $g \neq 1$ we have

$$0 = \lambda_0(g) + \lambda_1(g) + \lambda_2(g) + 3\chi(g).$$

At $g = d_1$, the three one-dimensional characters all take the value 1, so $\chi(d_1) = -1$. At a and a^{-1}, their values sum to $1 + \omega + \omega^2 = 0$, so $\chi(a) = \chi(a^{-1}) = 0$. Consequently, the complete character table is

Representative	1	(12)(34)	(123)	(132)
Class size	1	3	4	4
λ_0	1	1	1	1
λ_1	1	1	ω	ω^2
λ_2	1	1	ω^2	ω
χ	3	-1	0	0

As a check, the last row has squared norm

$$\langle \chi, \chi \rangle = \frac{1 \cdot 3^2 + 3 \cdot (-1)^2 + 4 \cdot 0^2 + 4 \cdot 0^2}{12} = 1.$$

□

5 Induced representations Throughout this section, G is a finite group, H is a subgroup of G, and all representations are finite-dimensional over $\mathbb{C}$.

5.1 Restriction and the construction of induction

If $N \triangleleft G$, a representation $\bar{\rho}\colon G/N \to \mathrm{GL}(V)$ gives a representation of G by

$$\rho(g) = \bar{\rho}(gN).$$

Its character is $\chi_\rho(g) = \chi_{\bar{\rho}}(gN)$, and it is irreducible if and only if $\bar{\rho}$ is, since their invariant subspaces coincide.

If $\rho\colon G \to \mathrm{GL}(V)$ is a representation, its *restriction* $\mathrm{Res}^G_H V$ is the same vector space with only the operators $\rho(h)$ for $h \in H$. Its character is the restriction of χ_V to H, denoted by $\mathrm{Res}^G_H \chi_V$. In the other direction, *induction* constructs a representation of G from a representation of H.

Definition 5.1 (Induced representation)**.** Let $\sigma\colon H \to \mathrm{GL}(W)$ be a representation, and choose representatives $t_1, \ldots, t_m$ for the left cosets of H in G, where $m = [G : H]$. Thus

$$G = \coprod_{i=1}^{m} t_i H.$$

For each i, take a copy $t_i \otimes W$ of W, whose vectors are denoted by $t_i \otimes w$ and depend linearly on w. The *induced representation* is

$$\mathrm{Ind}^G_H W = \bigoplus_{i=1}^{m} t_i \otimes W$$

with the action of G defined as follows. Given $g \in G$, write $gt_i = t_j h$ with $h \in H$ and define

$$g \cdot (t_i \otimes w) = t_j \otimes \sigma(h) w.$$

To check the representation law, suppose $gt_i = t_j h$ and $g' t_j = t_k h'$. Then $g'gt_i = t_k h' h$, and

$$g' \cdot \big(g \cdot (t_i \otimes w)\big) = t_k \otimes \sigma(h')\sigma(h) w = t_k \otimes \sigma(h'h) w = (g'g) \cdot (t_i \otimes w).$$

The identity acts as the identity, so the construction gives a representation of G.

Example 5.2 (The induced action in S_3)**.** Let $G = S_3$, $H := A_3 = \langle a \rangle$ with $a = (123)$, and $b = (12)$. Then $G = H \sqcup bH$, so we take $t_1 = 1$ and $t_2 = b$. We define a representation $\sigma\colon H \to \mathbb{C}^*$ by

$$\sigma(a) v = \omega v,$$

where $\omega = e^{2\pi i/3}$. The induced space has basis

$$v_1 = 1 \otimes 1, \qquad v_2 = b \otimes 1.$$

To apply the defining formula, write $gt_i = t_j h$ for $g = a, b$:

g	t_i	t_j	h
a	1	1	a
a	b	b	a^{-1}
b	1	b	1
b	b	1	1

Here $ab = ba^{-1}$ and $b^2 = 1$ give the decompositions in the second and fourth rows. For example,

$$a \cdot v_2 = b \otimes \sigma(a^{-1})1 = \omega^{-1} v_2.$$

Thus, relative to the ordered basis (v_1, v_2), the induced representation ρ is given by

$$\rho(a) = \begin{pmatrix} \omega & 0 \\ 0 & \omega^{-1} \end{pmatrix}, \qquad \rho(b) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The element a preserves each coset summand and acts by the corresponding scalar, while b exchanges the two summands.

Example 5.3 (The induced action in A_4)**.** Retain the notation a, d_1, d_2, d_3, V_4 and χ from Example 4.6. We construct a representation with character χ by inducing a one-dimensional representation of V_4. Let $U = \mathbb{C}$ carry the representation $\sigma\colon V_4 \to \mathbb{C}^*$ given by

$$\sigma(1) = \sigma(d_1) = 1, \qquad \sigma(d_2) = \sigma(d_3) = -1.$$

This is the nontrivial character of $V_4/\langle d_1 \rangle \cong C_2$ composed with the quotient map. Take the left coset representatives $1, a, a^2$ and let ρ be the induced representation on

$$V = \operatorname{Ind}_{V_4}^{A_4} U, \qquad u_j = a^j \otimes 1 \quad (j = 0, 1, 2).$$

The element a acts by

$$a \cdot u_0 = u_1, \qquad a \cdot u_1 = u_2, \qquad a \cdot u_2 = u_0.$$

For d_1, the defining formula gives

$$d_1 \cdot u_j = a^j \otimes \sigma(a^{-j} d_1 a^j)1 \qquad (j = 0, 1, 2).$$

Since $a^{-1} d_1 a = d_2$ and $a^{-2} d_1 a^2 = d_3$, the matrices in the ordered basis (u_0, u_1, u_2) are

$$\rho(a) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \rho(d_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Taking traces, and using $\rho(a^2) = \rho(a)^2$, yields

g	1	d_1	a	a^2
$\chi_\rho(g)$	3	-1	0	0

These elements represent all four conjugacy classes, so $\chi_\rho = \chi$. Thus induction gives a concrete realization of the degree-3 irreducible character in Example 4.6.

5.2 The induced character

Proposition 5.4 (Induced character formula)**.** *Let $\sigma\colon H \to \mathrm{GL}(W)$ be a representation with character φ, and let χ be the character of $\operatorname{Ind}_H^G W$. Choose left coset representatives $t_1, \dots, t_m$, where $m = [G : H]$. For every $g \in G$, we have*

$$\begin{aligned}\chi(g) &= \sum_{\substack{1 \le i \le m \\ t_i^{-1} g t_i \in H}} \varphi(t_i^{-1} g t_i) \\ &= \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1} g x \in H}} \varphi(x^{-1} g x).\end{aligned}$$

In particular, $\chi(1) = [G : H]\varphi(1)$, and $\chi(g) = 0$ if no conjugate of g belongs to H.

Proof. The action of g sends the summand $t_i \otimes W$ to the summand corresponding to $g t_i H$. Only summands with $g t_i H = t_i H$ contribute to the trace. This condition is equivalent to $t_i^{-1} g t_i \in H$, and on such a summand the action is $\sigma(t_i^{-1} g t_i)$. Taking the trace gives the first formula.

Every $x \in t_i H$ has a unique expression $x = t_i h$ with $h \in H$. For these elements,

$$x^{-1} g x = h^{-1}(t_i^{-1} g t_i) h.$$

Thus either the whole coset contributes to the second sum or none of it does. Since φ is constant on conjugacy classes of H, each of the $|H|$ elements of a contributing coset gives the same value $\varphi(t_i^{-1} g t_i)$. This proves the second formula and its stated consequences. □

Corollary 5.5 (Induction from a normal subgroup)**.** *Suppose $H \triangleleft G$, and let $t_1, \dots, t_m$ be left coset representatives, where $m = [G : H]$. If φ is the character of W and χ is the character of $\operatorname{Ind}_H^G W$, then*

$$\chi(g) = \begin{cases} \displaystyle\sum_{i=1}^{m} \varphi(t_i^{-1} g t_i), & g \in H, \\ 0, & g \notin H. \end{cases}$$

If additionally $\varphi(x^{-1} h x) = \varphi(h)$ for all $x \in G$ and $h \in H$, then $\chi(h) = [G : H]\varphi(h)$ for $h \in H$.

We denote this character by $\operatorname{Ind}_H^G \varphi$. The same formula defines $\operatorname{Ind}_H^G \varphi$ for any class function φ on H.

Example 5.6 (Permutation and regular representations)**.** Let $\mathbf{1}_H$ denote the trivial representation of H. In $\operatorname{Ind}_H^G \mathbf{1}_H$, the group permutes the basis vectors $t_i \otimes 1$ exactly as it permutes the left cosets $t_i H$. Hence

$$\operatorname{Ind}_H^G \mathbf{1}_H \cong \mathbb{C}[G/H],$$

where $\mathbb{C}[G/H]$ is the vector space with one basis vector for each left coset. Its character at g is the number of cosets fixed by g. When $H = \{1\}$, this is the regular representation $\mathbb{C}[G]$ of Section 3.

5.3 Frobenius reciprocity

For a finite group K, write

$$\langle \alpha, \beta \rangle_K = \frac{1}{|K|} \sum_{k \in K} \alpha(k)\overline{\beta(k)}.$$

The subscript specifies which group is used in the inner product.

Theorem 5.7 (Frobenius reciprocity)**.** *Let φ be a class function on H, and let ψ be a class function on G. Then*

$$\left\langle \operatorname{Ind}_H^G \varphi, \psi \right\rangle_G = \left\langle \varphi, \operatorname{Res}_H^G \psi \right\rangle_H.$$

Proof. By the induced character formula, the left-hand side is

$$\frac{1}{|G||H|} \sum_{g \in G} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \varphi(x^{-1}gx)\overline{\psi(g)}.$$

For each fixed $x \in G$, substitute $h = x^{-1}gx$, so that $g = xhx^{-1}$. Since ψ is a class function on G, this gives

$$\begin{aligned} \left\langle \operatorname{Ind}_H^G \varphi, \psi \right\rangle_G &= \frac{1}{|G||H|} \sum_{x \in G} \sum_{h \in H} \varphi(h)\overline{\psi(xhx^{-1})} \\ &= \frac{1}{|H|} \sum_{h \in H} \varphi(h)\overline{\psi(h)} \\ &= \left\langle \varphi, \operatorname{Res}_H^G \psi \right\rangle_H. \end{aligned}$$

□

If W is an irreducible representation of H and V is an irreducible representation of G, Frobenius reciprocity says that

$$\text{multiplicity of } V \text{ in } \operatorname{Ind}_H^G W = \text{multiplicity of } W \text{ in } \operatorname{Res}_H^G V.$$

Indeed, the character inner products on the two sides of the theorem give these multiplicities, which are nonnegative integers.

Example 5.8. Let $H = A_3 = \langle a \rangle \leq S_3$, where $a = (123)$, and set $\omega = e^{2\pi i/3}$. Let $W = \mathbb{C}$ with H-action $a \cdot z = \omega z$, so its character is $\theta(a^k) = \omega^k$. The standard representation V_{std} has character values $(2, -1, -1)$ on $1, a, a^2$, so

$$\operatorname{Res}_H^{S_3} \chi_{\text{std}} = \theta + \theta^2.$$

Since θ and θ^2 are distinct irreducible characters of H, Frobenius reciprocity gives

$$\left\langle \operatorname{Ind}_H^{S_3} \theta, \chi_{\text{std}} \right\rangle_{S_3} = \langle \theta, \theta + \theta^2 \rangle_H = 1.$$

Thus V_{std} occurs once in $\operatorname{Ind}_H^{S_3} W$. Both representations have dimension 2, hence

$$\operatorname{Ind}_{A_3}^{S_3} W \cong V_{\text{std}}.$$

5.4 A calculation in S_3

Example 5.9 (Induction from C_3 to S_3)**.** Let $G = S_3$, $H = A_3 = \langle a \rangle$ with $a = (123)$, and $\omega = e^{2\pi i/3}$. For $j = 0, 1, 2$, let W_j be the one-dimensional representation of H with character

$$\theta_j(a^k) = \omega^{jk}.$$

For $j = 1$, the induced action was constructed explicitly in Example 5.2. Compute $\operatorname{Ind}_H^{S_3} W_j$ for each j.

Solution. The subgroup H has index 2, so every induced representation has degree 2. Take left coset representatives 1 and $b = (12)$. Since $b^{-1}ab = a^{-1}$, Proposition 5.4 gives

$$\big(\operatorname{Ind}_H^{S_3} \theta_j\big)(a) = \theta_j(a) + \theta_j(a^{-1}) = \omega^j + \omega^{-j}.$$

No transposition is conjugate to an element of H, so each induced character vanishes on transpositions. Using $1 + \omega + \omega^2 = 0$, we obtain

Representative	1	(12)	(123)
Class size	1	3	2
$\operatorname{Ind}_H^{S_3} \theta_0$	2	0	2
$\operatorname{Ind}_H^{S_3} \theta_1$	2	0	-1
$\operatorname{Ind}_H^{S_3} \theta_2$	2	0	-1

Comparing with Example 2.7, the first row is $\chi_{\text{triv}} + \chi_{\text{sgn}}$, and the other two rows equal χ_{std}. The trivial character therefore gives

$$\operatorname{Ind}_H^{S_3} W_0 \cong \mathbf{1}_{S_3} \oplus \operatorname{sgn},$$

while the two nontrivial characters give

$$\operatorname{Ind}_H^{S_3} W_1 \cong \operatorname{Ind}_H^{S_3} W_2 \cong W_{\text{std}},$$

where W_{std} is the standard representation of S_3. In particular, induction need not preserve irreducibility, and distinct irreducible representations of H can induce isomorphic representations of G. The character induced from either θ_1 or θ_2 has squared norm

$$\frac{1 \cdot 2^2 + 3 \cdot 0^2 + 2 \cdot (-1)^2}{6} = 1,$$

which also proves its irreducibility directly.

We can recover the decompositions using Frobenius reciprocity. Restricting the three irreducible characters of S_3 to H gives

$$\begin{aligned} \operatorname{Res}_H^{S_3} \chi_{\text{triv}} &= \theta_0, \\ \operatorname{Res}_H^{S_3} \chi_{\text{sgn}} &= \theta_0, \\ \operatorname{Res}_H^{S_3} \chi_{\text{std}} &= \theta_1 + \theta_2. \end{aligned}$$

The last identity follows by comparing the values $(2, -1, -1)$ at $1, a, a^2$. Thus $\operatorname{Ind}_H^{S_3} W_0$ contains the trivial and sign representations once each and contains no standard summand. For $j = 1, 2$, the multiplicity of the standard representation is

$$\big\langle \operatorname{Ind}_H^{S_3} \theta_j, \chi_{\text{std}} \big\rangle_{S_3} = \langle \theta_j, \theta_1 + \theta_2 \rangle_H = 1,$$

while the trivial and sign multiplicities are zero. □

Exercise 5.1 (Induction from a nonnormal subgroup)**.** Let $H = \langle (12) \rangle \leq S_3$, and let ε be its nontrivial one-dimensional character. Compute the values of $\operatorname{Ind}_H^{S_3} \mathbf{1}_H$ and $\operatorname{Ind}_H^{S_3} \varepsilon$ on the three conjugacy classes of S_3. Use Frobenius reciprocity to show that their irreducible decompositions are $\mathbf{1}_{S_3} \oplus W_{\text{std}}$ and $\text{sgn} \oplus W_{\text{std}}$, respectively.

Solution. Put $s = (12)$, so $H = \{1, s\}$ and $\varepsilon(s) = -1$. Both induced representations have dimension $[S_3 : H] = 3$. Their characters vanish on 3-cycles, since no 3-cycle is conjugate to an element of H. The only conjugate of s in H is s itself, and $C_{S_3}(s) = H$. Thus, for $\theta = \mathbf{1}_H$ or ε, Proposition 5.4 gives

$$\left(\operatorname{Ind}_H^{S_3} \theta\right)(s) = \frac{|C_{S_3}(s)|}{|H|}\theta(s) = \theta(s).$$

Consequently, the induced characters have values

Representative	1	(12)	(123)
Class size	1	3	2
$\operatorname{Ind}_H^{S_3} \mathbf{1}_H$	3	1	0
$\operatorname{Ind}_H^{S_3} \varepsilon$	3	-1	0

The irreducible characters of S_3 restrict to H as follows:

$$\begin{aligned} \operatorname{Res}_H^{S_3} \chi_{\text{triv}} &= \mathbf{1}_H, \\ \operatorname{Res}_H^{S_3} \chi_{\text{sgn}} &= \varepsilon, \\ \operatorname{Res}_H^{S_3} \chi_{\text{std}} &= \mathbf{1}_H + \varepsilon. \end{aligned}$$

The last identity follows by comparing the values $(2, 0)$ at $1, s$. For an irreducible representation V of S_3 with character χ_V, Frobenius reciprocity gives

$$\left\langle \operatorname{Ind}_H^{S_3} \theta, \chi_V \right\rangle_{S_3} = \left\langle \theta, \operatorname{Res}_H^{S_3} \chi_V \right\rangle_H.$$

Since $\mathbf{1}_H$ and ε are distinct irreducible characters of H, the multiplicities of the trivial, sign, and standard representations are $(1, 0, 1)$ for $\theta = \mathbf{1}_H$ and $(0, 1, 1)$ for $\theta = \varepsilon$. Therefore

$$\operatorname{Ind}_H^{S_3} \mathbf{1}_H \cong \mathbf{1}_{S_3} \oplus W_{\text{std}},$$

while

$$\operatorname{Ind}_H^{S_3} \varepsilon \cong \text{sgn} \oplus W_{\text{std}}.$$

□

5.5 Application: a group of order 21

Example 5.10 (A character table from induction)**.** Let G be the group generated by the permutations

$$a(x) = x + 1, \qquad b(x) = 4x$$

of $\mathbb{Z}/7\mathbb{Z}$. Compute the character table of G by inducing one-dimensional representations of $H = \langle a \rangle$.

Solution. The generators satisfy

$$a^7 = b^3 = 1, \qquad b^{-1}ab = a^2.$$

Every element has the form $a^r b^s$, with $0 \le r \le 6$ and $0 \le s \le 2$. These 21 elements are distinct, since $a^r b^s$ sends x to $4^s x + r$. Thus $|G| = 21$, $H \triangleleft G$, and $G/H \cong C_3$.

Conjugation by b^{-1} sends a^r to a^{2r}, so the two nonidentity conjugacy classes in H are

$$\{a, a^2, a^4\}, \qquad \{a^3, a^5, a^6\}.$$

Moreover,

$$a^{-r} b a^r = a^{3r} b, \qquad a^{-r} b^2 a^r = a^r b^2.$$

As r varies, these give all seven elements of Hb and Hb^2, respectively. Conjugation preserves these cosets because G/H is abelian, so each is a conjugacy class. The representatives and sizes are therefore

Representative	1	a	a^3	b	b^2
Class size	1	3	3	7	7

Let $\omega = e^{2\pi i/3}$. The three characters of G/H give the distinct one-dimensional characters

$$\lambda_j(a^r b^s) = \omega^{js} \qquad (j = 0, 1, 2).$$

To obtain the remaining characters, put $\zeta = e^{2\pi i/7}$ and define

$$\theta_k(a^r) = \zeta^{kr}, \qquad \psi_k = \operatorname{Ind}_H^G \theta_k \qquad (1 \le k \le 6).$$

Using the left coset representatives $1, b, b^2$, Corollary 5.5 gives

$$\psi_k(a^r) = \zeta^{kr} + \zeta^{2kr} + \zeta^{4kr}, \qquad \psi_k(g) = 0 \quad (g \notin H).$$

In particular,

$$\operatorname{Res}_H^G \psi_k = \theta_k + \theta_{2k} + \theta_{4k},$$

where subscripts are read modulo 7. The three characters on the right are distinct, so Frobenius reciprocity yields

$$\langle \psi_k, \psi_k \rangle_G = \langle \theta_k, \operatorname{Res}_H^G \psi_k \rangle_H = 1.$$

Thus each ψ_k is irreducible. The restrictions of ψ_1 and ψ_3 have disjoint sets of irreducible constituents, so $\psi_1 \ne \psi_3$.

Set

$$\alpha = \zeta + \zeta^2 + \zeta^4, \qquad \beta = \zeta^3 + \zeta^5 + \zeta^6 = \overline{\alpha}.$$

We have found five distinct irreducible characters, one for each conjugacy class, and their table is

Representative	1	a	a^3	b	b^2
Class size	1	3	3	7	7
λ_0	1	1	1	1	1
λ_1	1	1	1	ω	ω^2
λ_2	1	1	1	ω^2	ω
ψ_1	3	α	β	0	0
ψ_3	3	β	α	0	0

The degrees satisfy $1^2 + 1^2 + 1^2 + 3^2 + 3^2 = 21$. Also,

$$\alpha + \beta = -1, \qquad \alpha\beta = 3 + \sum_{r=1}^{6} \zeta^r = 2,$$

so the two values α, β are roots of $t^2 + t + 2$. □

6 Arithmetic properties of characters Our main goal is to prove that the degree of every irreducible complex character of a finite group G divides $|G|$.

6.1 Algebraic integers

Definition 6.1. A complex number α is an *algebraic integer* if it is a root of a monic polynomial with integer coefficients:

$$\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_1\alpha + c_0 = 0, \qquad c_0, \ldots, c_{n-1} \in \mathbb{Z}, \quad n \geq 1.$$

Lemma 6.2 (An integer-matrix criterion)**.** *A complex number is an algebraic integer if and only if it is an eigenvalue of a square matrix with integer entries.*

Proof. An eigenvalue of an integer matrix is a root of its characteristic polynomial, which is monic with integer coefficients. Conversely, suppose α satisfies the equation in the definition. In the standard basis $e_1, \ldots, e_n$, define an integer matrix C by

$$Ce_i = e_{i+1} \quad (1 \leq i < n), \qquad Ce_n = -\sum_{j=0}^{n-1} c_j e_{j+1}.$$

The nonzero row vector $r = (1, \alpha, \ldots, \alpha^{n-1})$ satisfies

$$rC = (\alpha, \alpha^2, \ldots, \alpha^n) = \alpha r.$$

Thus $C - \alpha I$ is singular, so α is an eigenvalue of C. □

Proposition 6.3. *If α and β are algebraic integers, then so are $\alpha \pm \beta$, $\alpha\beta$, and $\overline{\alpha}$.*

Proof. By Lemma 6.2, choose integer matrices A, B of sizes m, n and nonzero column vectors u, v such that

$$Au = \alpha u, \qquad Bv = \beta v.$$

The vector $u \otimes v$ is nonzero, and

$$(A \otimes B)(u \otimes v) = \alpha\beta(u \otimes v),$$
$$(A \otimes I_n \pm I_m \otimes B)(u \otimes v) = (\alpha \pm \beta)(u \otimes v).$$

These matrices have integer entries, so the same lemma proves that $\alpha\beta$ and $\alpha \pm \beta$ are algebraic integers. Finally, $A\overline{u} = \overline{\alpha}\,\overline{u}$ proves the assertion for $\overline{\alpha}$. □

Proposition 6.4. *A rational algebraic integer is an integer.*

Proof. Write $\alpha = r/s$, where $r, s \in \mathbb{Z}$, $s > 0$, and $\gcd(r, s) = 1$. Substituting into a monic equation and multiplying by s^n gives

$$r^n + c_{n-1}r^{n-1}s + \cdots + c_1 r s^{n-1} + c_0 s^n = 0.$$

Hence $s \mid r^n$, and coprimality forces $s = 1$. □

Corollary 6.5 (Arithmetic of character values)**.** *Every value of a complex character of a finite group is an algebraic integer. In particular, a rational character value is an integer.*

Proof. If g has order n, then $\rho(g)^n = I$, so each eigenvalue of $\rho(g)$ is an nth root of unity. Its trace $\chi(g)$ is a sum of these roots, hence an algebraic integer by Proposition 6.3. The last assertion follows from Proposition 6.4. □

6.2 Class sums and character degrees

The space $\mathbb{C}[G]$ from Section 3 becomes an algebra by extending $e_x e_y = e_{xy}$ bilinearly. For a conjugacy class C, its *class sum* is

$$z_C = \sum_{x \in C} e_x.$$

Conjugation permutes C, so $e_t z_C e_{t^{-1}} = z_C$ for every $t \in G$. A representation $\rho\colon G \to \mathrm{GL}(V)$ extends linearly to $\rho\colon \mathbb{C}[G] \to \mathrm{End}(V)$ by setting $\rho(e_x) = \rho(x)$. By linearity, $\rho(z_C) = \sum_{x \in C} \rho(x)$.

Lemma 6.6 (The scalar associated with a class sum)**.** *Let $\rho\colon G \to \mathrm{GL}(V)$ be an irreducible representation of degree d with character χ, and let $c \in C$. Then*

$$\rho(z_C) = \frac{|C|\chi(c)}{d}\,\mathrm{id}_V,$$

and the scalar $\frac{|C|\chi(c)}{d}$ is an algebraic integer.

Proof. For every $t \in G$,

$$\rho(t)\rho(z_C)\rho(t)^{-1} = \sum_{x \in C} \rho(txt^{-1}) = \rho(z_C).$$

Schur's lemma gives $\rho(z_C) = \lambda\,\mathrm{id}_V$. Taking traces yields $d\lambda = \sum_{x \in C} \chi(x) = |C|\chi(c)$.

To prove integrality, consider left multiplication by z_C on $\mathbb{C}[G]$:

$$L_C(e_t) = \sum_{x \in C} e_{xt}.$$

Its matrix in the basis $\{e_t : t \in G\}$ has integer entries. By Proposition 3.2, the regular representation contains a subrepresentation isomorphic to ρ. On that subrepresentation $L_C = \sum_{x \in C} \rho_{\mathrm{reg}}(x)$ acts by the scalar λ. Hence λ is an eigenvalue of an integer matrix and is an algebraic integer by Lemma 6.2. □

Theorem 6.7 (Divisibility of character degrees)**.** *If χ is an irreducible complex character of a finite group G, then $\chi(1)$ divides $|G|$.*

Proof. Put $d = \chi(1)$, and let $C_1, \dots, C_r$ be the conjugacy classes, with representatives $c_1, \dots, c_r$. By Lemma 6.6, each number

$$\lambda_i = \frac{|C_i|\chi(c_i)}{d}$$

is an algebraic integer. So is $\overline{\chi(c_i)} = \chi(c_i^{-1})$, by Corollary 6.5. Row orthogonality gives

$$\sum_{i=1}^{r} \lambda_i \overline{\chi(c_i)} = \frac{1}{d} \sum_{i=1}^{r} |C_i|\,|\chi(c_i)|^2 = \frac{|G|}{d}.$$

The left side is an algebraic integer by Proposition 6.3. Since $|G|/d$ is rational, Proposition 6.4 shows that it is an integer. □

6.3 Applications to character degrees

Corollary 6.8. *If $|G| = p^n$ for a prime p, then every irreducible character degree is a power of p. In particular, every group of order p^2 is abelian.*

Proof. The first assertion follows from Theorem 6.7. If $|G| = p^2$ and an irreducible degree d were greater than 1, then $d \geq p$. Together with the trivial character, this would give $1 + d^2 > p^2$, contradicting the sum-of-squares formula. Thus every irreducible character has degree 1, and the sum-of-squares formula gives p^2 such characters. There are therefore p^2 conjugacy classes, each a singleton, so G is abelian. □

Example 6.9 (Degrees for a group of order 12)**.** Suppose that $|G| = 12$ and G has four conjugacy classes. The possible irreducible degrees are $1, 2, 3$: they divide 12, and their squares are at most 12. If n_i is the number of characters of degree i, then

$$n_1 + n_2 + n_3 = 4, \qquad n_1 + 4n_2 + 9n_3 = 12.$$

Subtracting gives $3n_2 + 8n_3 = 8$, whose only nonnegative integer solution is $n_2 = 0$, $n_3 = 1$. Hence the degrees are $1, 1, 1, 3$, as in the character table of A_4.

Exercise 6.1 (Determining degrees)**.** Suppose that G has order 18 and exactly six conjugacy classes. Determine all its irreducible character degrees and the multiplicities in its regular representation.

Exercise 6.2 (Rational character values)**.** Let $C_n = \langle g \rangle$ and $\zeta = e^{2\pi i/n}$. Consider the representation with $\rho(g) = \operatorname{diag}(\zeta, \zeta^{-1})$. Show that if $\chi_\rho(g) = 2\cos(2\pi/n)$ is rational, then it belongs to $\{-2, -1, 0, 1, 2\}$.